

Uh ma'am good morning aside. So we have officers from the line. So once ready we can start off. Uh first you can have the problem statement and then after in case of any specific queries uh sequentially we can start sir we'll start that uh the asking of the queries from the startup side. Thank you. So just give me one minute. I'm just checking if recording is happening. It is start. It has started. Just uh cross check. Once again good morning uh all of you. Welcome uh for the first outreach of uh uh Aditi 4 and disk 14. We have nodal officers available and today we will be covering uh four of Aditi four challenges and two of disk 14 challenges. Uh first the nodal officers will uh brief about uh each challenge and then we will take up Q&A for uh one by one for the respective problem statements. Uh I'm I'm going to tell what all uh problem statements we will be discussing today. We will be discussing 155 mm terminally guided munitions, uh, cognitive electronic warfare, programmable prefragmented and proximity fuse ammunition, humanoid for combat or humanoid auto system. These are the four problem statements that we will be covering. And from disk 14, we will be covering manportable alien system with ESM capabilities. Integrated drone management system for monitoring decision support system. uh once the technical outreach that means the Q&A uh related to the problem statements are done I will be covering the general out uh general outreach. So um I request uh the uh nodal officer from RT5 can please brief about the uh problem statement of Aditi 4 that is 155 mm terminally guided munition. The problem statement is flashed on the screen so you can take up. >> Uh thank you. Very good morning to all. The uh as the uh challenge is very name is clear, we looking for a 51 terminal guided munition. Uh it may be terminally guided or precision guided that is uh open to all. The aim of the whole thing is that it should have a precision guidance. It should be able to hit a target with precision with a CP of less than 10 uh 10 m. Uh where this technology uh has a challenge is the miniaturization of the circuitry that is involved in the sensor heads and if or the uh any glide mechanisms are being introduced onto the projectile. Now you can request just you can scroll down the problem statement. So the expecties are 5 ohm TGM which should be compatible with all three calibers. Even if it is not compatible with all three calibers, we are okay with one caliber initially and as a separate case we will then go for spiral development of the other caliber systems are there. What would be required in the indigenous guidance solution where GPS ins or labor or a hybrid for both of them all capability would be required. We would also request that the system whatever is being developed at present can be initially compatible with only planes but at a later time again we can see as a spiral thing that we can go in for compatibility with for firing in high altitude terrain. uh the requirements are very large. We already have these ammunition scaled at the rate of nearly 1.5 rounds per first line and 75 rounds per second line. uh the quantities required uh if not if I talk about today is somewhere around at the scale of around uh 10,000 but in another five five years the requirement of quantities is going to be somewhere around uh 60 to 80,000 rounds so the yes exists that is what I would have got to Yeah, >> thank you so much sir. Uh we will be flashing the second problem statement uh for Aditi 4. It is cognitive electronic warfare. uh the startups who are joining later uh I am reiter reiterating that uh uh we will be covering problem statement by the nodal officer first and then one by one for the specific problem statement we will be taking the Q&A uh so I request uh the nal officer from PMO suraj to please take up and uh uh over to you sir uh ma'am so for suraj there are two problem statement that is cognitive electronic warfare system as also the man cultable adding system with ESM cap capabilities. So the call option is the option of manate you will give both the problem statement together right so you uh right so you can uh uh so initially you can please explain regarding Adi four problem statement

that is cognitive electronic warfare system and after that we will be covering from disk 14 portable man portable alien system with ESM capabilities. Yes sir. Correct. >> Good morning ma'am and all the fellow panel. >> Uh the problem statement is uh AW as seen in the slide AW has grown uh to become an important concept in the military operations. It consists of three divisions support attack and protection for either offensive or defensive operations. Uh the system uh basically consists of antenna, receiver, processing unit, database to provide the means to intercept, identify, analyze and and localize. These are all theoretical. Uh so what we were looking is a a a receiver that will process, analyze and categorized as either threatening or non-threatening associated with the actions to be taken. Basically these data uh this will have ma'am can you scroll down a little >> right sir further more sir >> yeah uh that's all this uh this challenge can be effectively addressed with the cognitive technology which is use of artificial intelligence which will sense adapt and self-learn environment challenges and possible interference and of a feedback based decision making mechanism to intelligently deploy the counter measures. Then they have got the so this is what is required uh for our we will speak as we progress in the u in the conference that's all we can >> right you want me to move to the disk 14 problem statement sir >> yes ming system >> yes sir so here on the I have flashed the problem statement from disk 14 that is manportable alien system with ESM capabilities. Uh you can take over sir. Basically the radar reception system these fall short of a diverse and evolving threats. There are a lot of new generation radars which are being deployed and the the requirement of a portable land system which can provide the ESN capabilities including the broad frequency coverage. The frequency coverage that we are looking is up to 40 GHz. uh it includes uh frequency coverage accuracy uh in the direction finding the DFA as well as the location finding using two or three uh such measures or we uh if it is on a platform we can use the same thing to to geolocate a particular uh radar. The proposed system aims to address the need by delivering a robust scalable now it has to be scalable modular solution that enhances electronic surveillance and threat detection capabilities. The uh can you scroll down a little? Not too much. Yeah. The the present limitation is of uh uh frequency. the the frequency that we have got is a very narrow frequency. However, the new generation radar which uh which are being used nowadays are going in the higher bands. These radars not only for the ground base, they are also being used for uh the the missions. So this is what we are looking for. Further we will discuss why we will be discussing in the as we go ahead. Thank you ma'am. >> Thank you sir. >> Uh next problem statement that uh we will be uh discussing is Aditi four problem statement programmable prefragmented and proximity fusion ammunition. sir you can uh take up. >> Okay. The uh next problem statement that we every looking for is the smart emission. Once I talk about smart emission, it has to be programmable. It has to be prefragmented and it has to have a proximity to engage aerial targets which uh now we can see every day on our screens in the not only the Gulf region but also in the uh Ukraine Russia campaigns. So uh the existing guns L70 guns are currently firing only two types of ammunition. One is the proximity ammunition and second is the high explosive that is on a direct impact. So we are looking for an ammunition which not only has the proximity but also is programmable to increase the heat tolerability. Uh so the ammunition there are three basic requirements. One is the proximity, second is the programmable and third is the prefragmented. The shell has to be prefragmented. uh so uh the existing L7 guns do not have any programmer do not have any uh associated components. So the as part of this challenge there will be a requirement of developing a smart ammunition a programming system and integrating it with the existing L70 guns in the phase

one. As of now India does not have any such solution but in the world over there are two companies uh two different types of solutions which are available. One is by the VA systems, second is by similar and third is by all. So VA and the the similar both these companies have got 3P brand name or the classification of the ammunition and as far as all go it is ahead class of ammunition. So uh there is a lot of reluctance in by these companies to share this technology with India and uh we feel that it is a must for countering the threats the air defense or the various drones threats that we are engaging in the uh current environment. >> Uh right sir thank you sir. Uh and now we will be discussing the last problem statement. Uh and after that we will open the Q&A session. Uh correction that is the second last uh we will be discussing now Aditi four problem statement humanoid for combat or humanoid for humanoid auto system. I have flashed the problem statement on the screen. Uh nodal officer from infantry 7 will take it up. Sir over to you. Uh so ma'am it's a challenge which is jointly launched by the enemy as well as infantry. So uh we can take input both and uh then last that is uh you take a bone management system. So after that the director will give out that also >> uh sir sir you would like to take on right now or to give out the problem statement. >> Yeah I'll take on this statement. Thanks sir. Uh good morning ladies and gentlemen. Uh today's challenge addresses a critical operational gap which uh reducing soldiers exposure in high risk high altitude and combat environment. Our soldier operate in extreme terance under seaway physical stress and the problem statement why it matters because soldier carries 20 to 25 kg of load in harsh conditions impact endurance and mission effectiveness. The exposure to direct fire ambush IDs extreme weather and oxygen constraint which results in increased injury, reduced operational cycles and long-term health impact. But what are the key takeaways for this problem statement? We must offload the most dangerous tasks to machines. The challenge definition is in brief is to develop an indigenous combat capable humanite system with high dexterity and mobility, autonomy with human in loops control and capability to operate alongside or ahead of soldiers. It must be modular and software upgradeable, adaptable to Indian terrain and doctrine and scalable in cap capability which may be a stage was development. The top technological domain which we are looking for from the industry are robotics and actuation to give high torque rugged actuators for human motions sensors and controls in the robot. We are looking for a multi-session fus sensor fusion for balance navigation perception and a IML integration for reinforcement learning of gate generation stability and decision making. The expected outcomes probably stage wise is first for the deployable system. Next is fully functional combat humanoid system platform capable of assisting the soldiers replacing replacing soldiers on high-risisk missions and testing and certification requirement which is his last stage of the prototyping. Further capability expected are stable movement on uneven icy inclined terrain humanike dexity weapon handling and equipment manipulation operational modes may be fully autonomous or semi-autonomous. The current gap which we are looking which are we have presently is there is no indigenous deployable human right combat system exist and the global solution are not tailored for the Indian conditions with this uh this is not just a robotics project it is about serving a soldier's life saving soldiers life and enhancing combat effectiveness and building strategic autonomy that is the brief of the uh this PDS. Thank you. uh uh request if you can just do for the uh integrated drone management system also. >> Yeah. Yeah. We go ahead with this. >> Right. >> Uh so for the management system ma'am can you just show this slide please? >> Yes. Yes sir. I'll uh I'll flash the next problem statement uh that is uh disk 14 problem statement integrated drone management system for monitoring decision support system. >> Yes, thank you >> sir. >> Uh so

in this project we are here today to engage with innovators startup and industry partners on a critical operational requirement of the Indian army which is to develop an integrated drone management system for monitoring and distance support. The nature of warfare is changing rapidly. The battlefield today is data driven, fast-paced and multi-domain. Drones are no longer standalone assets, but they are force multipliers. The problem statement per se. At present, multiple drones are deployed across the technical battle area. However, these systems operate in silos. Video feeds are isolated and localized and there is no unified system for aggregation and analysis. This leads to the fragmented operational picture and limits the commander's ability to take timely and informed decisions. What we lack today is a centralized command and control system, a common operating picture across multiple drones and a real-time fusion of aerial and ground data. The vision of the required system as per the PDS is to connect all drones tethered or untethered stream real-time video and sensor data provides a single window operational dashboard and most importantly the system must enable AI driven decent support to commanders at all levels. The key functional expectations will be centralized monitoring, AI enabled analytics, secure and resilient communication, interoperability, drone and mission management which includes fleet health monitoring, mission planning, task allocation control, etc. And what we expect from industry is it should be a scalable solution. It should be field ready sub-system with a strong AI capability and engagement approach like it has to be a prototype development field testing and with user unit iterative refinements and scaled deployment. Ladies and gentlemen, this is an opportunity to be part of the building a next generation operational capability. We are not just looking for the vendors. We are looking for partners in innovation. Thank you so much sir. I have a question. Thank you sir. Uh can I ask a question now? >> Wait, we will be uh starting the Q&A session and now now we are going to begin the Q&A session. Uh first I would like so that it will be easy to timestamp the uh recorded session. So I would like the innovators who have questions related to the Adi 4 problem statement that is 155 mm terminally guided munition. I'll flash the problem statement and innovators who have uh questions related to this problem statement can please uh ask their questions. >> Uh good morning everyone. This is Shaavag from Blackstone Tactical. Uh can I ask the question? >> Hello. >> Yes. Uh please go ahead. Uh sir, when we talking about this terminal guided ammunition, >> is there any uh startup who has questions related to 155 mm terminally guided munition? >> Exactly ma'am. This is what I'm asking. Can I go ahead? Hello. Can I go ahead? >> Uh yes. Uh Mr. Shen, please go ahead with your query. There is there is just one question I would like to ask like u we here we are talking about the fuse with a fmg. So are we also looking? >> Uh if there is if there is no question related to the first problem statement of aduti that is 155 mm terminally guided munition. I'll move to the next problem statement. Questions related to the problem statement that is cognitive electronic warfare. Uh startups can ask their questions. Officer is with us. Is there somebody from IDO IDEX they are not able to hear the point the question answer session is going on and in between interruption is there so can somebody talk to them on phone and guide >> just a bit sir uh uh ma'am >> I request the startups to raise their hand and uh uh we will unmute the startup and take up the question >> ma'am I'm in between the question can I go ahead Hello. >> Uh request everyone to just hold on with the I I guess they are not able to hear us. So I'm just uh giving a call to them. Just hold on. Everyone request just hold on. >> Sure. >> Uh yes. Uh there is some uh technical issue. I am asking the technical person to join. Uh I believe the startups are asking questions but I'm not able to uh listen to them. Wait for >> Hello. Am I audible? >> Shahavas was asking us a question

but his voice is not clear. >> Can I go ahead separate now? Okay, Mr. Yes. >> So your voice there in the voice but I'm just asking a question as for my understanding you are asking about a fuse with a wing with a fin so that it can change the trajectory. Now my only question is are you looking for laser guided tool and if yes then what kind of laser guidance will be there? So GPS I can understand um it can have a AI system but what about the laser guidance? what kind of laser we are used to we are going to use to guide the thing >> the first outset I would want to say that I'm not looking with for a fuse with a fin >> a f becomes a smaller system it is it comes to the aspect of a course correction fuse or a precision guidance scale which is much smaller and not is not that technical in detail and the accuracy is much lesser. We are looking for a complete projectile, a 5 mm projectile which has a fuse which can be guided and the fin the fins on the body which once they deploy are able to move with the guidance from that onto the target. As far as the laser is up, we are not specifying what bandwidth of the laser there because there are many commercial lasers also available. uh and when the uh strengthening of the system is done the first aspect that has to be seen and developed upon is the guidance is the ability to guide and control the system at that spin rate that in case of any question we can get to or we can meet in person for the clarification. Uh yes uh this out sessions are particular for that particular reason. In case you have some queries which comes up later you can write to the and they can be compiled and shared with us. >> So sir I have a question. uh I wanted to understand for this particular problem statement what would be the mode of engagement like as startups I think during the development of the prototype we would we would need assistance with let's say the actual projectile which we can modify and test our solutions with so is it something that will be provided by the ADB or we are supposed to build prototypes with arbitrary projectiles and then transfer it to the real projectile >> uh s I see it uh This is going to be more of a a consortium based thing where everybody who has the specializations will have to join in together and create a shell or a design and uh there are a good amount of vendors in the market who are actually doing this uh today where the projectiles are being designed and being prepared. We can as part of the project if you want access to all projectile that we have that will have to be checked and confirmed because as of now the inventory that we have is very very limited so that cannot be confirmed as of date. >> Got it. >> Hello. Uh so I have another question. Uh just wanted to know is there a range parameter uh to this as well? So the range parameters are 85% of the stand uh fuse standard uh H shell fuse for various calibers. So for a uh 52 caliber gun a standard 85% of this standard H shell comes to be somewhere around 24 to 24 25 26 km. However, if it is fin deployed and it is GPS line is guided with the glide mechanism, it should be able to in around 40 to 50 km or so. Uh those aspects can be deliberated at a later stage and we can definitely help you on those aspects later. Got it. Thank you. question. >> Hello. Am I audible? >> Uh yes, you already >> uh so my question is regarding uh the DIC 14 uh that swarm drone coordination uh uh challenge. Uh in this we have already developed uh the ground station. Uh uh just a minute uh right now we just taking a query with respect to 15 TGM that is terminally guided. So later on we take on others. So anybody who have any further more queries they can go ahead. >> Oh sure. >> May I ask sir? >> Yes please go ahead. Please >> sir. Uh there are two three small questions. One is is laser compulsory or because most of the projectiles like Excalibur they only work on GNSS. Uh to answer you laser is not compulsory. You can get us either a G GPS based solution or a laser based solution or solution in conc. >> Okay sir. Uh so the second is I think uh a project with similar name was floated in make 2 also is the specs are same for that and this one or they are different sir. >> So the make 2 project is a different project. Uh the specs may or

may not be the same. Uh as and when what are the options that the industry gets to us we will work on that and then finalize the specs upon that. >> Understood sir. Sir, the last question is sir uh if we want to make something like an excaliber then the entire shell will have to be redesigned whereas if we make a course correction fuse then it becomes much easier. Would you take both the solutions or you are uh wanting only one type? >> So you can submit for the course correction field only. However, this prevent case right now is for uh this and as it is uh uh as I see your name, you are already involved in a similar case of a similar course fuse. >> Yes sir. Okay. So no in case uh no more queries related to them. So can we move to the next uh the problem statement that is of cognitive electronic warfare system. uh any uh queries uh regarding cognitive electronic warfare system or shall we move to the next one? >> I have one question sir goes for you. >> Yes. Yes. Please go ahead Mr. Raju. >> Okay. This is regarding uh the direction finding. So is this similar to what RWR would give like a sector wise direction or it should be like precise one as far as the uh as far as the direction finding system is concerned it will give one system will give a direction using two or three system you will be able to locate uh the pinpoint location with a fair degree of accuracy. See that is what is required. >> Okay. >> Yes sir. I have a question as well. Sir, my question is regarding the uh frequency range of the system like what we are covering uh as uh mentioned earlier uh in the man portable system we are looking up to 40 GHz. So is this the uh is it the same for this system as well? Uh as far uh this is concerned uh we are on electronic uh cognitive electronic war uh the LPI signals which determine how much uh bandwidth of the system which we need to consider or does this not include that >> it uh it do include that uh in the LPA signals the bandwidth specific bandwidth will be shared once we get the technical space vetted from all the stakeholders but yes it will be included. >> Thank you sir. >> Hello sir I have a question. Can I >> sir? >> Hello sir. >> Yeah please go on. Go ahead. >> Hello sir. My name is Rohit Chanas. I'm from Huso Space Private Limited, Guati. I have a question. Uh so it was mentioned in during the presentation that uh uh the frequencies that uh you're looking for is in the gigahertz range. But for uh terrain like northeast because of the hilly region VHF frequencies would better serve for uh like a tactical uh offline intelligent uh communications network. So for that purpose if a system is designed for optimized for the various frequency so would it be applicable for this uh category or are you specifically looking for a like a system which is optimized for gigahz range because we are working on a very high frequency range uh system currently that's why I have this question if you could >> Mr. Rohit we are looking for a solution which will be deployed in all across uh terrain. Obviously it will have a VHF component also as well as it will go up to GHz range where uh there are a lot of ammun uh emission happening and it will 18 to 40 GHz also that we have seen. uh although I I understand this point that the VHF as well as HF frequency are being used much in the models but there are places where uh the the higher frequency has got its employment. So we are looking for such a system which has got all these facilities. So sir uh if uh the presentation this project if it is made in the uh VHF array and we have uh opponents uh make uh uh like arrangements for uh like gigahash range in future generations. So would it be acceptable in that case? Uh no it will not be accepted uh even in this case uh that we are we have not specified that we only require in the VHF range we require in a very high uh bandwidth from VHF till it goes to the microwave frequency ranges. >> Okay sir. Okay sir. Thank you sir. >> Actually I'm Kai from MMRfic. I also have a question. May I ask? >> Yes sir, please go ahead. >> Uh my understanding of this project is that you already have your EW systems. What you want to add is some cognitive features in the processing part of it. Okay. And whereby you have defined what

you want in uh direct finding etc. Is that understanding right >> sir? U it it is actually if you see sir there are certain uh we we do have some systems of AW and we are also in the procurement as well as looking for some more EW uh systems >> in future as far as now is concerned uh we are also looking for a solution based approach in which AI or ML will be used a cognitive approach for the existing system will be used uh uh but that we are discussing with the OEM who has provided this particular system. As of now what we are discussing now is a project a complete project which has got a in-built cognitive EW system a complete system complet uh comprising of uh all the things that that that has been stated in the problem statement uh and a total solution sir. >> It's uh so that's that's why I'm asking that clarification. Uh is it the complete hardware also to be supplied along with that including? >> Yes. >> Yes. >> Uh good morning sir. Uh myself is Pravin Reddi from Astra Microwave. Uh may I ask a question on regarding alien system and portable alien system? >> Right now uh so we just focusing on the problem statement two that is cognitive then after we move to so that we can discuss. >> Okay. May I ask a question? Hello. Please go ahead. >> Okay. Uh in the problem statement it is mentioned that various EW sensors. So what all EW sensors we are considering for this problem statement? Uh Mr. Kartik, we are looking for uh EW systems in terms of uh HF, VHF, uh drones, satellites as well as uh there are other u uh non-com domain also. So all these type of system consist of PW. It also consists of jammers related to this detection as well as jamming. >> Okay sir. Thank you. >> Um sir I have one question Koteswar here. Uh this is regarding EA side. uh are you only considering uh the noise or baras jamming or are you also considering the RGP or VGPO based uh advanced jamming for the prototype >> Mr. uh potential. We are looking for the old type of jamming techniques also as well as the new type of uh cognitive jamming also in this particular system because the radio set system will be handling will have both types of uh systems the barrage jamming as well as all the new generation type of jamming the responsive jamming sort of thing. >> Okay. >> Thank you sir. Uh sir I have one more question. So what all sensors we are using for correlation and corabbration separately. >> Uh Mr. Karthik we are looking for a solution from your side. As of now we don't have much uh uh >> not we we don't have this particular thing existing. So these are our uh requirements that we are looking for. Presently we don't have any system which has got this cognitive ability. >> Hello. >> Okay sir. Okay. Okay. Thank you. >> Hello sir. Can you hear me? I had a question regarding terminal cat in munitions. Uh uh we have actually moved on a second. Uh so in case you have query you can write to diio and they can compile and then they can share because uh right now we just discussing the cognitive electron the officer or the the subject m expert they have uh actually moved out. So you can just write to they will compile and then give the reply. >> Got it. Write to DIO's mail. >> Okay. >> Can you share email ID on which DIO email ID? We need to >> uh we'll do that. will inform the DIO to share uh the link of this recording as well as Zoom in where the startups can uh give their queries a solution. I had couple of questions on the automated system. Uh one was regarding fix are you initializing this to be a single system or can it be a model and uh am I sir am I audible? >> Yes. Uh yes ma'am you are audible now. Right. So uh so I'll be sharing uh email id where you can post your queries and I'll post them further to the respective model officers regarding uh sharing of uh this uh recording. I won't be able to share it to the respective email ids that startups are dropping in. We will upload this video on our YouTube channel. It will be uploaded with all the u with the timestamps and Q&A session uh the the the full video. So uh you can stop sharing your email ids uh over here. >> Hello. >> Uh uh sir uh because I I was not in may I know uh till where we have done the Q&A session? Uh right now we have done with the first one that is TGM. Uh we

are on second one cognitive electronic warfare system. So right we are taking queries two. >> So startups if they have any queries any more queries related to cognitive electronic warfare can post their queries and I'll be sharing the email id. Uh uh I'm just typing it. This is Zahir from Big Bang Boom Solutions. Uh couple of questions on the cognitive EW system. Uh the first one was uh are you visualizing this system to be a single system or can we also visualize it to be uh specially distributed to be able to get the location data? Mr. Zah, we are looking for a system which is a integrated system. If you are talking of specific, it will have control block DF system jamming uh jamming of VHF uh HF as well as uh jamming of cellular frequencies. It can have a cellular interception also satellite also as well as non-com and and UAV capability. So this cannot come in one system. It will be a integrated sort of system consisting of all these types of entities. >> Understood. Uh yes. Uh second question is regarding the AML part which you were talking you be giving us some access to some of the signals as part of known threats on your library or uh this is up to us to build a threat library sir. Uh they are uh very specific if you uh give us in the query we will look uh how much we can share uh about our database. Uh so this I cannot comment now but once we get the query we will discuss uh uh amongst ourself and we will uh help you out in this. >> Got it. >> Thank you. >> Yes. ADB uh and um just I wanted to ask if you're talking about uh CW cognitive electronic warfare so in this uh threat library uh from where it comes because when we are talking about cognitive uh then it will be open you will be receiving you will have an AI model uh that will be reading out and then you will be doing doing autonomous which we are talking about the advanced system. So what I why I'm talking I'm I am first of all I am from the PI Institute of Science. So I'm talking about that the specifications when we bring it out to the startup that should be very clearly known what is the starting and where it limits. So um that's why I'm raising this question when we are talking about the cognitive electronic warfare which is today the end thing. So I wanted to ask from Mr. Jahid that from where it came to your mind that there will be a library provided by to you sir as part of the uh problem statement uh there is a section that talks about a knowledge base with prefed uh threats. So that was the basis of my question sir. Anyway that I will leave it on to the ADB but why I'm bringing out this point that the other startups when we are talking about the advanced cognitive part so that the the major part in this is that your readout the where where receiver is there that will be receiving you'll have an AI element that ele AI element will work it out on that and then you will generate your own whatever you are going to do that because that is not that it will be always jamming only. So please look at it. Uh if India has come out with such a good objective so it should be you who should be because the good word which I saw from the ADB the partnering in the innovation the word. So please look at it that you also partner to make this uh better for our services that was my submission to you. So because um the more you ask the questions um which is not there then it puts in other startups mind that it is something different directions they will move through. If I am correct uh ADB can endorse. If I'm wrong uh they can correct it. >> Right. Thank you so much sir. So lots of innovators are posting that uh for the query email id uh I am spelling it out. You can post your uh queries onto this ID but I have a request to give time to the nodal officer to respond to those queries because those queries will come to me. I'll share them the same day. They will uh they they will need time to reply to those queries and then I will forward those uh uh queries the replies to those queries to you. I'm spelling out the email id. You can note it down. It is Gorov Dio 1989 Golf Alpha Uniform Romeo Alpha Victor Delta India Oscar 1 1989 at the rate zoho mail.in In you can note down this email id. uh you can uh send your queries and I request you to please write in the subject the problem the name of

the problem statement and disk or aditi it is it will be easier for us to uh forward the query to the respective nodal officers and uh uh this is my humble request to all of you that that please don't wait for the 11th hour 4th May is the last date please please apply in time don't wait for the 3rd May to come and then apply Okay. Uh if all the queries related to yes please repeat the email id. >> Okay. The email id is gor that is golf alpha uniform romeo alpha victor dio delta India oscar 1 1989 at the rate zoho mail.in. I have posted it on the chat also. You can uh uh take email id from uh the chat also related to the YouTube uh related to the recording of this session. It will be posted on the YouTube channel. You can take reference from that. >> Ma'am, are you going to answer the queries that have been put in chat? >> Yes sir, please. >> Excuse me. And that is cooking. So now we are moving on to the third problem statement that is the programmable field fragmented and proximity fuse amission that is for 40 mm. So uh uh so anybody who have any startup who have these query with respect to uh adi problem number three they can uh ask their queries. Thank you. >> Uh sir I have only one question. This is Shahas from Blackstone Tactical. So we are already working on this system. So there is two partition the programmer. Now the problem is that do we also have to develop this uh range distance finder because ahead is going to explode based on that distance. So are we going to get information from your radar or your existing system or we have to design that detection target locking system too. Sir And we have some upgraded L70 guns in which the muzzle velocity radar is already there. So you may not uh design the MV radar. However, uh we would like to have a composite solution from your side so that we can upgrade the unupgraded L70 guns which that means the entire solution which is required to fire the smart ammunition from the L70 gun should be designed uh by the uh innovation. So you mean uh also the detector also like and everything that kind of thing. >> I have not understood the detector part of it sir. >> Yeah like I mean there will be a system that will detect and then the fire will calculate and then it will program the fuser. >> Okay. So uh we uh have the electrooptical system on our guns which has got the LRF. So LRF may not be required to be put onto the gun but the uh other components of the smart ammunition like the ammunition itself, the programmer and the integration integration as part of the integration the uh ICD and ids of the existing UFCs system which is fitted uh would be provided. Sure. >> Thank you. Please. >> Hello. Can I ask my question? >> Yeah, please ask your question. >> My name is uh my name is Sudar. I'm from Le defense system. uh uh the the uh gun system uh I think the expectation is that there should not be any modification or very minimal modification to the L70 you know gun right. So the shells are normally uh spin stabilized. So if it uh it has to be spin stabilized or is it okay uh if you use uh even a smooth board you know gun and using some alternative uh stabilization method. >> Okay sir. So uh what we are looking at how do you make the uh system as as smart is by incorporating a programmable fuse. Rest of the system remains the same that is the cartridge the propellant and the shell. shell will also be the preformed uh shell should be there and so we are not expecting any uh modifications to the gun because it's a it's a rifled gun. It's not a smooth bow gun. So the spin stabilization will continue and uh we are not going to change the barrel. So the ammunition has to be designed to be able to be fired from the existing guns. Okay. Thank you. Sir, I had one question uh with respect to the fire control architecture that you cleared earlier. Now you mentioned that a EO fire control system is already present in the L70 guns but not implemented completely. So the solution that you're looking at is only from a uh programmable fuse, ammunition and a programmer or does it also have to include a third component a fire control system? No. So in the phase one we are only looking at the smart ammunition smart programmable ammunition

not the electrooptical fire control system. EFCS systems are separate programs which are on through which we will upgrade the guns. However, we have adequate number of guns for fitting the upg the the smart ammunition programmer and fire it fire the smart ammunition. So no we are only looking at the smart ammunition and its programmer. >> Okay. What realtime engagement data is available at the gun level for the fuse programming sir from the prior control system? what all can be made available uh the real time engagement data it could be something like range velocity uh target tracking data what all will be available even electrooptical fires control system can pick up what the datas are available that includes the uh 3D uh target data 3D target data that the bearing range and the uh elevation of the target >> as well is we have the mother velocity radar for finding out the muzzle velocity of the round which has been fired four data points would be available. >> So this it is assumed that this fire control system also includes a ballistic computer also. >> Correct sir. Once we talk about electrooptical fire control system then it our understanding is uh clear that it consists of a day camera, night camera, LR along with the LSD computer. >> One last question are we talking about L7 existing So uh we will be allowed to attack a 6 in item at the end of the baron. Obviously I mean that is that will that is how we are going to program. >> So you may have your solution sir. What I have seen is uh that before the round is loaded through the reach end and before the round is fired the program inductive programming happens at the reach end itself and not at the muzzle end. So uh yes the uh you need to have some programming system inductive or some system by which the round this program it has just been fired. >> Okay. So any question sir uh >> sorry to introduce uh sir I have a query sir. regarding uh sir can we uh fire control rate can we change or not sir uh >> I have not understood your query >> sir fire control rate can we change is that possible if we go for uh fire control rate sir if you want to change >> so the rate of fire of the gun is generally up to 300 rounds per minute uh I I don't see any reason for changing the rate of fire of the gun. It will remain the same uh because the programming will not or the inductive program will not affect the rate of fire of the gun. >> Okay sir. And also sir uh for uh uh for barrel sir can we go for internal grinding? I don't see any requirement for internal grinding of the barrel for using the smart ammunition and uh we don't recommend any changes in the barrel or the main gun itself. >> Okay. >> Sir, what are the quantities tentatively that we are looking at for the ammunition as well as for the programmer for the for this problem statement? So initial quantities that we have given out in the that is the proto development the quantities are adequate to make a business sense uh not be able to give out the open domain but it is adequate quantities are there along with the requirement of ammunition not the current but in future as well as in the global scenario I'm sure you would have seen the the current conflicts which are on people are looking for ammunition gun ammunition of this particular caliber and this particular type engagement cost. So yes, both Indian army as well as world. >> Sir, can I ask one more question? >> Please go ahead. >> Yeah. In in the warhead, right? uh is there any specification in terms of number of fragments that needs to be know let us say that you know if uh the pellets are let's say uh of 1.5 mm or 2 mm know diameter right is there any number right so many in a thousand fragments or little >> okay so the current emissions which are available worldwide ranges from 2 to 3 up to 3 to 5 mm uh preformed. So you may have your own solution but I recommend that you use the existing prefragmented rounds which the industry has already developed. Only thing that needs to be needs to be developed is the proximity and the programming along with the programmer. So it it would make more sense if I had been you to develop to develop the uh fuse part of it and collaborate for the uh pre-fragmented part of it to

develop the electronic portion of the fuse the programmer and integrate it with the existing shell and the cartridge case to be able to fire it from the existing guns. >> Okay. Thank you sir. Are there any questions with this problem statement? Okay, we will move to the next problem statement that is humanoid or combat humanoid auto system. Startups who have questions related to this can uh uh ask their questions and I have dropped the YouTube link uh to the channel uh where we will be uploading this uh recording. >> Good morning sir. Uh my name is Faridan. I'm from General Autonomy. Uh can I go ahead and ask the question? Yes, please go ahead. Yes, please. Sure sir. >> So uh my question is to the technical representative from the EME. Uh sir in the problem statement uh there's a section for project outcome where it is said that there are stages 1 2 and three that will result in three deployable products exoskeletons. Sir exoskeletons is a very different technology from humanoids. Are we expected to first build out exoskeletons and then go towards humanoids later or was it a reference to a teleoperable humanoid? uh which got called out as exoskeletons like basically what is the actual expectation that is there at these three stages sir >> uh Mr. So far actually uh when we did the uh interaction with many industry partners uh here we found out that the combat robot robot uh can be can be developed only through the various stages in which one of the first stage was the ex development of exoskeleton and if you have any other solution available with you we that is you're always welcome >> understood sir thank you sir >> good morning I have a similar question uh the uh the requirement calls for a humanoid and the reference to exoskeleton. Are we only considering the humanoid or are we uh also looking at other UGVs and other uh physical form factors? Uh sir uh we are looking only for the commodized robot platform uh which uh depicts a humanoid movements as of now for this problem statement. >> Thank you. >> And and sir uh the problem statement talks about the autonomous uh robots uh but I believe like the first set of solution would probably involve some form of teley operations. Is is that part of the solution that is acceptable? Uh because the full autonomy will be built out in multiple phases and I think one of the phases that we need to encounter first is telly operations. >> Can you repeat the questions again please? >> So this question is about the autonomous humanoid which is to be deployed. I wanted to understand if if the initial solution which involves teley operation is that is that something which is acceptable and present as part of the road map or do we need full autonomous solutions from get go >> okay we are looking for a full autonomous system however uh uh the specific question part of the specific question you can just we'll revert back to you can share with the uh DIO and we'll revert back to the same question again thank Sir, this is Bhanu from General Autonomy. Uh, what is the amount of indigenous content you're looking at in humanoid? Because certain components are not available from Indian players. So, we'll have to make this ourselves. So, clarity on indigenous content percentage would be useful for us. So uh as of now we are looking for make in India uh category 2 and if it is category one it is always welcome. >> All right. Thank you sir. Uh do we have more questions? >> Hi from the >> I have a question regarding the payload capacity. Uh in the talk it's mentioned the soldiers carry 20 to 25 kg of payload. Uh so what is the payload capacity expected from the humanoid? Yeah, we were looking for uh 25 kg and above capacity so that it is uh with the some uh some platform if it has to be utilized in some different operational environment the efficiency does not reduce. So anything above 25 kg Yeah, good morning. Uh I'm Raju this side from scient in the technology domain. Uh the uh question is like uh advanced battery and power systems uh uh is to be used for high density and field su energy. So wanted to understand the advanced battery what is the configuration you are looking at? Uh

sir uh exact conflict won't be we won't be able to tell you right now but what I request you can share the question is to the we'll revert back to this. Thank you. >> Thank you sir. >> Hello can I can I take one question? >> Yes please. Yes please. >> I'm Anil from ADSL. Uh this is regarding uh your problem statement number one T72 autonomous armored fighting vehicle platform. Can I take can anybody take a question on this? >> No, we don't have nodal officer for this problem statement. We are not covering this problem statement today. >> Okay. So who may be the nodal officer if you can tell me. >> Uh just give me a moment. I'll tell you when we will be conducting the outreach. We have just today only we have started the outreaches. Is it the first problem statement for disk 14? Uh this is uh >> this is I think your this is4 >> uh uh is it uh conversion of tank 72 into autonomous armored fighting vehicle. >> Correct. >> Yes. It will be on uh >> 27th sir it will be on 10th. It will be on 10th of April. >> Perfect. Thank you so much. >> Hello. >> Hello. >> Uh sir, in the human problem statement, what kind of capabilities in terms of operations? What kind of operations do you want this human to conduct in terms of handling weapon or carrying load? What kind of operations we are looking at? Uh sir as the PDS uh clearly states that it should carry certain load that is 25 kg and above in any operation condition. Second it should carry uh this is one part of it and second is it can uh if required can be work alongside the soldier. He can carry the gun. it can patrol the particular area and be part of some uh team in which it will be controlled by the humans but uh the robot will part of the team. >> So I am asking like let's say there is some specific uh operations uh like digging for the mines and de mine all those things needs to be covered with this single humanoid. So the uh the use cases for this robot human robot maybe uh may vary from one portion one ter to another ter and the operation requirement. Uh but what we can do is we can share the exact terms and condition for him for a robot to work in the office and come back to you in this regard. Sure. Sure sir. >> Uh hi sir. >> Good morning. Hello. >> Yes, you are audible. You can ask your question. >> Yes. Yes, sir. So, sir, like uh there were some other problem statements in the same uh like autonomous uh sector like uh for humanoids, trucks and also there was a problem statement for uh robot dogs. uh does that uh require all common system or do we have any separate uh preferences or any condition for uh humanite robots as well? >> Uh this problem statement is also we will be covering another day. Uh >> no no no like uh for human right do we have any separate conditions for that or all three categories have similar path. Uh hello. >> So probably the all the category would be coming on the same category but uh uh the idx people will mean uh will tell you about this. >> Yes. If you are uh I I'm I'm talking to the innovator who has asked the question. If you are if if your question is related to uh the problem statement which has robbo which involves robbo that problem statement we will be covering on probably 28th of April. We will keep posting all the schedules on social media. You can keep following. We will put the problem statements that we are going to cover. The respective nodal officer who has the more knowledge about the problem statement will be uh joining us to explain more and uh reply to the Q&A. >> Okay. Thank you. Thank you so much. >> Okay. Do we have any more questions related to humanoid for combat? >> Yes sir. Yes sir. Sir Robots landitude hill sir. Uh question altitude hilly termin. Night robot. As of now we are working on the highitude mobility. Okay. Okay. Okay. Sure. Thank you sir. Um this is Sepati from Red Nuts. I have a question. Can I ask? >> Yeah please go ahead. Uh so uh in for humanoids uh there is uh there's lot of in loop training that is required along with the forces. So what is the plan for this? So for any operations let's say combat or otherwise. Uh there would be a closed loop training uh which would be required. What kind of support and plan uh is done for this? >> Uh yes you are right. uh the closed loop training would

be required to train the robots for the particular mission or the particular operational requirement. That uh part will uh be adjusted during the prototyping of the uh uh robot maybe in TRL 1 or TRL 2 level. Understood. Understood. Thank you. Hello. >> Yes, please. >> Uh, so I have the question regarding human rights robots that uh for particularly for the submission of this solution. What level of completion is required for the prototype? Is it the design stage? Is the idea stage for the basic prototype? We are looking we are looking for a final pro final product uh per se. We not looking for some prototype but uh the first prototype product should uh demonstrate the complete capability which we have put up in the PDS and or uh during the Q&A session which will be delivered bible back to you. >> Okay. uh after this Q&A session we will be covering the general outreach uh in which you will you will be able to understand how the process will be so maybe you will get your answer from there >> sure do we have any more questions related to humanoid for combat If we do not have any more questions, we will move to the disk 14 problem statement. Uh man, portable alien system with the ESM capabilities. >> I think uh ma'am just resume >> right. So >> uh if you can start with the drone management system we can we are already here. >> Yes sir. Yes sir. Yes sir. We so we will start Q&A session for the last following statement that is integrated drone management system for monitoring decision support system. Uh startups can ask their questions. Hello ma'am. So I had one question. So for this integrated role management system what is the uh computational requirement or the base station requirement that we are looking at? Is it supposed to be deployed at a headquarters level or is it supposed to be deployed at an operational level? Uh so sir uh deployment uh it is as of now what as for the PDS it is at the core headquarter level in which the complete uh integrated video or the output of the uh data output of the all the drone in the particular DBA tactical battle area would be fed to the core headquarter level. However, uh what we are forcing is that there would be some uh modules kind of a thing deployed ahead in the during the in the TBA which will be feeding it to the direct to the core headquarter centrally. >> Got it. Thank you. >> Hello sir, I am Vijendra from ADL. I have a one doubt like as the problem statement you have mentioned that you want the video for feed from the all type of drones available in the Indian defense system. So few of the drones which are have perpetual license for the communication system. So who who have the responsibility to provide the feed to the command and control system like are we are going to uh get the access or permission to put any modification in the system are the uh owner of the system is going to provide feed to us. uh your initially all the licenses will be correlated and then it will be verified at uh the by the manufacturer and then it can be uh promulgated to everybody. >> Hello this is Sanu from Axelron. Um can I proceed with the question? >> Yes, please go ahead. >> Yeah, thank you. So I understood like this will be deployed on a headquarters level but still I would like to know like what is the approximate distance we need to cover and what is the landscape meaning this will be extremely hilly area as well as a grounded area or the temperature meaning we should be withstand the extreme hot temperature with a also the low low temperature as well. Can you please help me on that like how this should work? >> It'll be combination of uh extremes itself because the uh drones can be the system can be made functional at both highly areas or the deserts. It can be heat areas, heat areas and planes as well. and an approximate uh size of this location mean how long this should cover. Uh sir, we'll uh we'll come back to this question uh this uh this question separately. However, just to give a fair idea uh uh the TB would be approximately 20 30 km ahead. It is not uh the correct picture though but approximate ballpark figure. However, the connection will be through the data communication network which is a

secure network from the TBA to the uh the next headquarter. >> Understood, sir. Thank you. >> And sir, one more question around this. Uh it I understand that the system has to be deployed at the headquarters level but uh do as part of the development process are we supposed to demonstrate the system working with the new set of drones or is it supposed to be deployed with whatever drone uh infrastructure the services already have. I'm asking this question because if we have to do this with the existing drones then we might have to uh embed them with some custom modules. Is that permissible as part of the process? >> Uh in this case both uh new systems and the old legacy systems both have to be integrated. >> Got it. So for legacy systems we are allowed to uh include a new communication module so that we can set up a communication with the HQ software or the ground control system that you're building. Right. >> Yes. Yes. >> Got it. I'm sorry. You can all you can give the question as well and in case any update is there we'll report back to you. >> Thank you. >> Good morning sir sir. >> Hi good morning. Uh can I take up a question? >> Yeah please. >> Yeah sir this is Anup from iotech world navigation private limited. So I just want you to understand when you talk about the control system like completely completely integrated solution to uh you know monitor and also control like it has to be a two-way communication let us say a person sitting from the command in headquarter he wants to control a drone flying in Rajasthan so he should be able to control also it's or it's only one direction that he want to view the videos and all those things the operation going on is it a two-way communication or only one one Sleep >> sir if s uh sir so basically uh uh it it is supposed to be two-way communication uh but it depends on the uh particular headquarter uh which is authorized to do the two-way or not. So it will be hybrid kind of a situation for this system. >> Okay. So for the communication uh you know I understand that you said that 30 km should be the range from where you want to monitor the drone or control the drone. So is the vendor uh uh you know uh can choose because there are various type of communications available nowadays. So is it uh possible for a vendor to select or we need to get the prior approval uh from uh headquarter before implementing such solutions. The type of communication that we need to use >> the communication should be secure basically as encrypted west communication framework. Uh this is the basic requirement. Answer. Uh second regarding the 30 km range it is not very you know conventionally it may may increase somewhere or may decrease somewhere. So we'll not fix in the range per se because as all the data is communicated through a data line which is secure data line. So we'll restrict we not restrict oursel with the limited coverage in this uh PDS. >> Okay. So in that case you are open for type of communication that a vendor intend to use to implement this solution. Right. Correct. >> Okay. Got it. Thank you. >> Good morning, sir. This is Re from Meridian Data Labs. Uh, can I ask one question, sir? >> Yeah, please, please. >> Yeah. Uh, so this just wanted to know sir, maybe I could have missed few of the your queries. uh one is uh uh one once we build the software uh are drones also part of like you already mentioned I believe there will be legacy uh drones and new drones right sir. >> Yes. Yes. >> So can we then alter the software uh of the drones to put our new software because we are building some particular software on the edge uh analytics. So can is there a possibility we can upgrade the existing software uh for the legacy uh drones that you already have or or maybe the new ones so that that can is possible. >> Uh we will not be able to comment on this one because the manufacturer of the drones should give you the privileges and permission to alter the software in the system. >> Okay. So at this point maybe we will not be able to comment on it yet. >> Yeah, we'll not be able to comment on that one. >> Okay. And and the AI capabilities that are uh sought out sought for uh it that has to be

done at the server or ground uh control station level or like I was mentioning on the edge itself on the drone itself. >> It can be on only three platforms. person. >> It can be on all the three platforms or either of the platforms. >> All the three or either of Okay. Okay. Maybe on the drone or at the server or JS GCS level, right? >> Got it. Got it. Understood, sir. Okay. Thank you for Hello, sir. Uh sir you just mentioned about the you the range can be anything maybe 20 kilometers 30 or more. So does it mean that you will be providing that channel or we will be using the existing channel which you guys already have or we have to develop that channel for data communication. Hello. >> Uh we'll have to check this one uh because uh the communication what you'll be using or the sorry the channel what you'll be using it has to be verified first. So we'll have to seek permission from the higherups and we'll come back to you. >> Okay. So as of now I understand that even that whatever that communication channel that also we have to build uh there is existing communication channel but whether you you'll be able to utilize that existing communication channel or you'll have to build your own uh that has to be sort uh because licensing the out the other person to utilize the existing channel has to be verified. So we need some time to check that one. >> Okay sir. Thank you. And uh sir during the development process will it be possible for ADP or AAD to help us with some of the existing infrastructure which exists so that we can design our solutions around it like give us a specifications of the kind of drones which are used or uh or the channels which are used communication channels Uh sir uh probably ADB uh will give back revert back to you and I request you to send the questioner in the email please. the students will be able to save this. Uh sir, can you please clarify again whether this will be a two-way communication or it uh uh having both monitoring and control or it will be just monitoring? Uh sir uh I'll repeat myself. This would be a hybrid model kind of a thing in which certain uh uh administrative powers will be given to some uh user to operate two-way and rest would be on a single way communication only. >> Okay. Thank you sir. Hello. I have a question. >> Yes, you ask your question. Just be slightly louder. >> Okay. So I was talking about the problem statement that we are talking about this integrated drone monitoring system and also the standalone DJ which at the end are providing a common uh solution at the end. So how can we differentiate between these two solutions because both at the end are providing a common solution where it's a two-way communication right when we talk about the GCS and integrated marketing system. Ma'am ma'am uh your voice is very low ma'am can you please repeat in a louder voice ma'am please? Uh yes sir sir was talking about the problem statement in VIC 16 and three which are a standalone GCS and integrated roing system as we talking about the communications these two problem statement are aiming a common solution at the end >> ma'am I think uh you are talking you you are talking about a different problem statement which we are not covering can you please repeat uh uh disk problem statement six did you 16 16 ma'am like integrated drone waring system and GCS which both the problem statements are aiming for the common solution at the end. >> A problem statement number 10 outreach will be on most probably will be on 23rd of April and uh uh 16 is the one that we are discussing right now. Yes. On the same problem statement, I was just trying to understand uh the solution part where a a standalone GCS would also be giving a two-way communication and you know to the headquarters while the integrated drone monitoring system would also be giving the same in terms of a solution. How can we put both of the solutions? Can we put both of these solutions together? treatment question >> uh at this stage I doubt you can put both the solutions in uh one solution. I think there are different problem statements so different solutions are expected. >> Okay. Uh do we have any more

questions related to this problem statement? If we do not have any uh immediate questions uh uh sir you can uh log off and uh I u I I I thank a lot the nodal officers ADB for uh joining the uh technical outreach and clarifying all the doubts related to the problem statement of the innovators. We will be uploading uh this uh uh recording uh on the >> ma'am the elent one has not been covered >> sir uh I think lent one u uh nitan sir told me that it is uh the nal office is from PMO sur and we already had another uh problem statement from the same nal diretory that is cognitive electronic warfare. So that time it was being covered. Uh if uh basically the internet connection Yeah. >> ma'am the lent is not covered. I am the officer. If any questions is regarding ma'am I uh it can be uh covered now ma'am. >> Okay. So uh if there are any queries related to the man portable alien system with ESM capabilities they can ask the questions now. >> Yeah. Can I ask the question please? >> Yes please. >> Yes sir. >> Yeah. So what is the frequency range the alent system should be capable of? >> Up to 40 GHz sir >> and what is the lower range.5 GHz.5 GHz? >> Yes sir. >> Okay. 0.5 to 40 GHz. Thank you very much. Uh sir can I can I ask a question? >> Hello? >> Yes sir. >> Uh sir uh you said the uh direction finding system is also required in the manportable system right? Uh sir just a second sir uh the u the requirement of the lowest frequency band is from 30 MHz not.5 30 MHz to 40 MHz >> I I corrected >> yeah that's my question sir okay thank you because uh to find out to to include the direction finding the lower frequency ranges should be >> sir sir uh please clarify 13 GHz to 4 GHz >> no no 30 30 MHz to 40 GHz. >> Okay. 30 MHz to 40 GHz. >> Yes sir. >> Thank you. >> And what is the size sir for this complete end system like uh weight weight of the system? Weight of the system we we call it man portable. So the weight of the system is not specified as of now because uh it should be of the order of less than 100 kgs I believe around 80 kgs or I think just a second I will just check. It should be mount on tripod, right sir? >> The antenna will be mounted on tripod. >> Soldier carries 20 kgs. >> So it's man portable sir. Basically it should be modular so that four soldiers can carry it and keep it somewhere. Now the weight is not specified sir. I have noted down this question. I will come back to you with the answer what weight we are looking for. So basically what you're saying is that it need not be uh operable on the move. It's transportable manportable not um manpack basis right? >> Yes. Yes. Man thinks it should be a a rack sort of system uh which can which can be very easily uh on a very uh fast time frame can be deployed uh although we're not looking for a solution which is on the no >> and do you look are you looking for a is there a moq that you have specified because triangulation is required then there's the integration aspects >> control systems all of that uh are you going to include all that or it's just one s one terminal that you're looking for. >> We are looking for in the pilot project we are looking for a minimum quantity of 10. The system consists of the frequency uh specified range. We have already told it will have a static as well as a non-portable uh design. The the direction finding systems to have uh accuracy of minus 75 dB of of operational range. We will have a basic analytical function as well as the radar fingerprinting system which is now essential for identifying complex threat. >> It will also the system will also have infra public quadrature recording facility for RF to which also include RF fingerprinting technology. And and this is all expected within the uh uh disk uh financial powers, right? >> Yes sir. This is all all >> Yes, sir. >> One question. >> Any any other question, sir? Sir, can I ask you one question? >> Yes, sir. >> Yeah. Uh, thank you for answering the question, sir. So, one thing I want to know is um since the coverage range coverage depends on the antenna height, how much at what height you can move the antenna. Uh, so do you have any restriction on what is the max height that you can raise the mast to? No sir, we don't have any restriction on the mast but uh looking at the

condition any mass about 10 to 15 uh I think 10 m uh the existing mass is I believe is around 5 m height >> okay >> vehicular base but this particular thing is not a vehicular base so from the ground if you have a height of 5 m we have not specify the height of the mast sir. Although I understand that the mast height comes with the RF figure length also. So that also has to be taken into consideration but we have not uh >> yeah the range height of the >> so basically if it is a portable thing we can uh put it on a place with a elevated platform that thing uh is not part of this project but we are working on that. Uh as far as the solution is concerned, we will see if it can be put on a particular uh uh tower or something of that elevated platform like a watchtower or some sentry tower or some something like that uh in forward areas. >> Okay. >> May we know the sponsoring directorate of this project please? It is PM2 sir. >> Thank you. >> Sir is since you wanted from 30 MHz sir uh is it possible uh that you're also looking at comment as well as we have already uh discussed comments uh when we had discussed the previous one. In this particular thing we are only looking for event >> only for event sir. So >> okay sir >> repeat the could you repeat the sensitivity value that you had given 70 or 75us. Thanks. >> Uh that that is the SFDR - 70 or >> that is the uh specifications the operational range -70 al also although we are working presently on - 85 dB is existing but uh the operational range of this particular portable system is we have specified as -70 dB. Uh sir for which parameter sir? >> The effective coverage of the antenna the uh >> sensitivity. >> Sensitivity. Exactly. >> Uh sir I have a related question to that. Uh the sensitivity number what you are giving is for a specific bandwidth. The IBW what we are looking at the incident yes it is uh it is IW for the entire band this is the minimum -70° uh it should uh give the sensitivity should be not uh uh it should be better than this >> the radars radars I mean they have different bandwidth right they come at like 100 MHz 30 MHz 40 MHz like that various bandwidths they are available so I'm wondering for what bandwidth we are we need to meet this minus 70 dBm number >> sir uh between the the range this is a radar ESM equipment >> correct >> looking in a frequency of 30 MHz to 40 GHz >> so we are looking for a um effective operational uh range of of sensitivity of -70 dB in the entire range of frequency coverage sir. No, that I agree sir I understood but what I'm asking is radars do not cover such a wide bandwidth even though they can be anywhere within the band but the bandwidth of individual radars will be shorter much okay so there are lots of radar starting from ground wave to ship base to to a airborne radar although groundbased radar works on low frequencies but the moment there are airborne radar it goes till beyond Xband and then there are new generation radar which are operating uh in more than 18 GHz. So that is why we are looking for a solution beyond 18 GHz till 40 GHz. >> Okay sir. Any further questions? Uh in the LN part in the man portable LN system. >> Uh sir this man portable you said can be vehicular mounted also sir. It can be we have specified it as a manportable thing but manportable means that a vehicle is not uh part of the solution. So if you give us a solution we will put it on a vehicle and we will so this vehicle is not part of this project. >> Yes sir. Because you said 10 m u that will be have not specified uh the height of the mask. I will come back to the specifications of the mask. As far as uh mask is concerned the we have not specified it in the uh in our QR sir and sir uh this uh you have mentioned the existing solution sir. >> Yes. So uh can we get a just a direction on what is uh H model or not the models but the functionality which you are looking for. We have a solution that is with the with the air force uh a company named R&Ds as well as Masloa they are using this they are giving the solution up to 40 GHz. Uh yes sir but that is not manportable sir u performance is uh like they've got a very small antenna so the DF uh DF accuracy is not there in that >> um are you talking about RNS solution or the meshova solution

>> sir RNS solution we can find some information sir uh as to what what is the direction that we need to go in but for the other company we are not aware I will not comment on their what we are getting in the in but we are also dealing with them separately regarding those the those issues uh what uh my uh in this presentation uh we are discussing about our requirement. So we have specified our requirements. Are you specifying a uh level of accuracy of geolocation that you desire? No sir, no we have not specified it. Excellent. >> Do we have more questions? Uh sir as far as uh uh just a second ma'am as far as your specific concern regarding the specifications we have not come to that uh level that uh to give a specified direction finding accuracy that we are looking as of now we have kept it a little generic the antenna height and the direction finding accuracy. So as and when we progress in this particular project we will uh let you know sir. Uh any specific uh questions related to other things can be addressed to the uh mail ID given by ma'am. >> Thank you. >> U ma'am secondly if I may just uh address this to IDEX. Uh this is a disk 7 1.5 uh uh it's a disk project ma'am it is 1.5 CR and uh will it be possible that in three three CR of u PDB that we'll be able to do it all these things >> and quantity is 10 >> uh sir uh the problem statements that have come from the Indian army are sent after the understanding that this projects have the max uh uh grant that is 1.5 and Aditi projects have the max grant of 25 CR. So it is not up to uh me to you know to change the challenge from disk to aditi and uh vice versa. So uh nodal agency has considered this and then they have put it in disk or uh uh the challenges have been categorized into disk or aditi. Secondly, the maximum uh grant is 1.5. Uh however, we do have projects which have the PDB of 4CR. Uh and the projects are running. Not that I'm suggesting that uh you go for this solution. I'm just giving you the information. >> Right. Ma'am, thank you so much. Uh also how do we find the details of nal officers for a particular problem statement or should we send an email to go dio? Yes, you can send the email to the email id that I have sent and uh uh we will forward the your queries to the respective nodal officers and we will revert back to you. That's why I'm saying that uh don't wait for the 11th hour. uh please uh post your questions and give nodal officer also a time of a week to revert you back and you work upon it subsequently. >> Okay. Thank you ma'am. Can you post the email address again in the chat box please? >> Yes sure. >> Thank you. uh also is the selection of the projects based on L1 or QCPS system. U meanwhile uh do we have any question? Uh let me answer this question. Uh Shakshi regarding L1 we right now the forms have been published where you given your tech proposals. Uh definitely there will be a uh due diligence being done on the costing also. So that is um the further instructions will be posted on the website also on this issue. So in the form we are not taking any budgeting part. We're just taking right now the tech tech proposals only but subsequently there will be uh the selected winners will be intimated uh on the go ahead of submitting in detail their proposal because that will be a gives a good understanding during your uh uh selection process uh initial selection process. So it's a two-stage selection. One is the first tech selection followed by financial selection. Thank you. Uh sir, I have one question related to the operating operation environment like high altitude whether it has to be working in the high altitude scenario as well. If so, what is the maximum height? It is not specified. Sir you can uh put your query s we will have an internal discussion and let you know. >> Okay sir. Thank you. >> Uh ma'am for all the challenges should we apply through IDEX website or for audit is a separate one and for other challenges separate? No, there is only one uh uh uh you can apply through the website of IDEX only irrespective of uh disori ma'am and all the queries uh which are being raised would be promulgated to all participants. >> Uh no the uh the the queries that will be emailed to us it is not possible to uh uh

send those queries to all the participants. it's it's it's practically not possible. So if you have any query, you can you can mail those queries to the uh email id that I have provided and uh I'll revert back with the response from the nodal officer. >> Great ma'am. Thank you so much. >> Do we have any technical query with respect to the problem statement? Do we have any more? Uh if we do not have any more technical ma'am >> as a concept if someone has asked a question we should all know the answer because uh you know uh for the sake of uniformity otherwise you will everyone will give you 200 questions. >> I agree uh I I I totally agree but it is practically not possible uh to uh email the response. Maybe uh you might need it, another might not need it. And it is very difficult to uh uh reply first receive the reply from the nodal agency and then reply to all the participants whose email ids are there. It is practically not possible. You the maximum that can be done is you you send me your query. I don't mind sending you reply as and when I receive from the nodal officer. >> Point point very well taken. This is an uh issue. We are trying to work it out on the website by having a query uh uh kind of a uh link wherein the uh the startups of the MSME can post a query and we'll try and answer there. uh let me check with the IT team what we are working do we have it uh uh available otherwise right now what Sakshi has mentioned kindly follow that thing give your query on the on the email we'll try and revert back to you on the email itself but this point is very well taken we need to have something which is which may be common for all yeah >> yeah sir if you don't mind can we suggest two options two solutions to Mhm. >> Uh so for now uh I heard uh that this is going to get uploaded on YouTube that means it is getting recorded. >> Uh option one is one of uh your uh team members uh if having the bandwidth can go through the recording and note down all the questions or else uh two to three among the attendees can take the owners who have noted down all the queries. they can just compile and give you >> sir that most welcome you you're free to compile and send it to us on the email we'll definitely have a try and have them answer it also yes that's a good suggestion >> a standard AI feature we we have a limitations you may be having AIS with you we have our own limitations here I mean there's a good suggestion from one of the participants you are free to uh give us the queries also on the email. We'll try and have a link for their answers as well as the queries on the website. >> Thank you sir. >> So we all have our login s in case uh >> uh excuse me excuse me please uh mean uh when uh I mean we can do this discussion uh once uh we are free now from the technical outreach. So let's uh uh free the nodal officers and then we can continue with with this uh discussion. Uh thank you so much for ADB uh uh thank you so much to ADB for onboarding on this uh uh technical outreach and bringing on the nodal officers uh whatever technical queries we had uh have been resolved and any more that will be coming I'll be sending them to the respective nodal officers email id sir you can log off. Thank you so much for joining sir. >> Uh thank you ma'am. Thank you everyone. Yeah, >> thank you. Uh, nodal officers can log off and and now we can begin with the general outreach and that time we can discuss about these questions. Uh, give me a moment. Uh, uh, I'll upload the uh, presentation that I have for the general outreach. Hello. >> Hello ma'am. >> Uh yes. Uh meanwhile uh if you have any general query you can uh uh ask the general queries uh when I'm loading the presentation. Yes please. uh and and please uh don't drop your email ids on this box because uh uh the recorded uh the the the recording will be uploaded to the YouTube channel. I have already shared the link to the YouTube channel on this chat box. >> Okay. This is Giran from Jaga opensource. I'm looking at the problem statement uh AI module for UIS for autonomous recognition admission targeting. Ma'am, >> wait a moment. I'll tell you when we will having the outreach with. >> Okay. >> Yeah. Can you repeat the problem

statement? >> Uh problem statement is AI module for UAS for autonomous recognition, identification and targeting. Wait a minute. Do you know the number of the problem statement? Which disc or >> statement number one ma'am? Uh Aditi. >> Pron statement number one. Aditi. >> Yes ma'am. >> Uh I don't the problem statement one from Aditi from army is different. I think you are referring to some other knowledge agency's problem statement. Okay, let me >> The problem statement number one from Aditi from army is conversion of tank 7 uh tank T72 into autonomous uh armored fighting vehicle platform. So this is >> ma'am he is asking about this problem statement one. Okay, wait a moment. A disk one problem statement is development of AI module for autonomic recognition identification targeting. will be having uh its outreach on uh 7 host. What is the date? >> 17th of April. >> 17th of 17th. >> Okay. >> Hello ma'am. This is from LIT lab. Ma'am, can you uh share the list of outreach dates? When >> I have Yeah, I have uh the dates but these dates are not final. That is why I'm not putting it on the public forum. Uh uh dates keep changing. That's why I'm not uh putting it on the uh public forum. But uh you can uh follow us on the social media. Uh we whenever we are posting the link to the outreach uh we are also posting the challenges that will be covered and also this time uh the link for the outreach is same. So this uh the link with which you have logged on all the outreaches irrespective like army, air force, navy, mha all the outreaches on the will uh will be conducted on the same link. uh you just have to uh know the schedule that when you're interest the problem statement that you are interested into when the outreach is there uh the the link is same >> uh which social media platform you are posting this thing >> all the social media platform we post LinkedIn Twitter uh Facebook and uh uh Instagram also >> ma'am we already have a login on uh DIO portal so can't we have a direct email uh about this instead of social media Yeah. >> Uh I doubt we can uh uh do that. Uh not to my knowledge that uh having your uh uh login and then this >> so so point taken it's a good suggestion. We will try and have it on the on the initial website page also. You need not to have a login. We'll work we'll tell our to upload the challenges there also. Otherwise every witness day we'll push the challenges coming for the for the coming week. Yeah. So that will be uh from that day itself we'll be start pushing on the social media the suggestion of yours on the website also we have taken it we'll we'll put it across. >> Thank you so much. >> M have a question regarding in the open challenge. >> Yeah I think you posted yeah regarding open challenges. Uh uh I can reply to the questions uh related to army challenges only. If you have your question related to army challenges, you can ask. >> Yeah, it is an army challenge ma'am which I applied around uh November 25. Uh still now there is a no response till today. It's nearly 4 to 5 months. >> Okay. So if you have applied uh on uh November 2025 uh you can wait. uh we are uh I have not received from the nal agency uh that the the the probable candidates for the HPSC uh uh you can wait for uh approximately one more month if you don't hear anything uh by the April end uh you can consider that it was not shortlisted this time you can apply again uh but by April end uh you might hear something >> but the idx portal reflecting the status in process ma'am Yes. Uh uh it shows it shows uh under pro under process because we are yet to close that uh uh OC that's why >> ma'am I have a question on similar lines >> does committee as commerce ma'am Are you there? So now I'll be uh beginning with the general outreach. I'll be telling you uh the what uh uh there there might be many uh innovators who already are part of IDEX. Uh those who are not part of IDEX, it will be very much useful for them and those who already are of the are of are the part of IDEX, it will be helpful for them also. So uh uh we are innovation for defense excellence and uh we come under uh defense innovation organization. Uh our aim is uh to uh to basically foster the innovation

ecosystem. We are there to merge the gap between defense ecosystem and uh the innovation uh startup mummy sector in uh in the private sector that is there. uh we are basically to bridge that gap between services and the startup ecosystem. Uh so DIIO is a section 8 company that is not for-profit company. We were formed under with the help of two DPSUs that is HL and Bell. Our mandate is to foster innovation in technology development in defense sector and to provide grant or funding and other support to carry out innovation. Uh by other support uh we mean uh assistance in uh uh in taking the matching contribution that is the the other part of the grant and also to assist the startup in understanding the needs of defense services. uh uh uh the DIO board consists of the secretary DP, secretary DD, R&D and chairman, DRDO, FEDS. We have uh representatives from uh uh HL and Bell. Their CMDs are are part of board. We have MD AM and NITO. NITO, ASDP and JSDIP. Uh so we were formed in 2017 as a section 8 company under DDP MOD and innovation for defense excellence is a scheme that offers financial support and uh grants in aid launched by DIO. We aim to foster innovation and technology development in the defense and aerospace sector. Uh we promote uh private sector led innovation and participation of startups, small business, individual innovator and R&D institutes in defense. We promote indigenization of defense technologies and reduce import dependence. Uh if I talk about IDEX grant, we have total three schemes. Uh the disk, the full form of disk is defense India startup challenge. The grant that we provide uh is maximum to 1.5 CR. Uh we have Aditi which is aiding development of innovation technologies with IDEX. The grant maximum is up to 25 CR. under open challenge which we call at uh call as OC uh the grant is maximum to 1.5C cr uh so there are different uh nodal uh there are different verticals under which you can uh submit your proposals uh the so motoral proposals by innovators come under open challenge and the uh problem statements that are launched by the vertical for example army air force navy DPSUs they come under disk or aditi ask. Yeah. May I take questions afterwards? >> Okay. Okay. Sure. >> Yeah. Uh so uh under uh so disk and uh the objective of DISK is to engage startup me and MSME to solve a specific defense challenge and in Aditi we are looking for uh looking at the accelerated development of critical defense technology. uh that is why the fund uh that we provide is uh uh going up to 25 CR. Uh so basically uh the grant that u the features that we have the grant is up to uh 3 million USD basically it is 25 cr in case of Aditi we promote co-creation and co-development by co-creation and co-development I mean that uh the innovation and the innovation is that we getting is coming from the startup and MSME and the uh the the suggestions and the uh problem statements are coming from services. Uh that's where we uh come with the co-creation and co-development. Uh we have dedicated person of contact for each project which will be the project officer from IDEX. Uh we give the matching contribution uh uh correction I I stand corrected. The matching contribution is reinforced by the committed startup or the small business. Uh I'll take up I'll go into more details regarding matching contribution later. uh we uh uh uh the the intellectual property rights that is IP is 100% uh the ownership of the innovator. We don't take anything from uh the IP and uh there is uh an assured procurement uh the MPQ is uh circulated. However, MPQ cannot be informed at this stage. uh uh when the startup the selected startup progress in the in the project that time we uh inform them about the MPQ uh when I'll come to the progress of the project the progress of the problem statement at this stage I would say that time you will be able to understand it better uh we have partner incubators and I network uh through which uh the partner incubators help in understanding the technology and handholding of the startup with respect to the development and I are there to assist the startup in securing the investors and uh the funds for the matching

contribution. Uh the IDEX challenge uh challenges uh span advanced defense technologies which include uh maritime uh domain AI VR the satellite based ISR smarting munitions unmanned autonomous system and advanced munitions. Uh this is important slide. So uh I'll take you to that how uh uh the challenges uh that we are discussing the problem statements we are discussing will progress. So uh the launch of the curation uh the uh the challenge curation we already have done and after that we have launched the pro uh problem statement at today as on date we are at outreach stage. After outreach you all will be applying and the last date for the application is 4th of May. uh we have a history we have not extended the last date any time so I request you all to not wait for the last moment please apply on time because uh and and please don't wait for the last day last day uh you might not know that what technical glitch happens so please apply on time because we have not done any extension so uh uh 4th May is the last date to apply after you apply we receive all the applications and depending upon the number of applications the nal agency takes time uh for uh sorting the applications and then uh uh it will uh the node agency sort out the applications and then we uh come to the HPSC. HPSC is the high power selection committee. Uh so now when I'm talking about HBSC we are at this stage. uh HP uh during HPSC uh the uh few uh few participants who have a few applicants I would say few applicants will receive uh the notification that they have been shortlisted for the HPSC. HPSC consists of various uh subject matter experts from various uh uh verticles from the industry, from the academia, from the nodal agency and they uh in HPSC the the startup the applicant is uh supposed to uh go into detail about their uh solution that they have proposed. Uh at this time they also propose the timelines, the budget that they are looking up to and uh other details. uh based on that the no the uh HPSC that is the committee decide that whom they want to shortlist and uh uh then we get a provisional winner. After getting a provisional uh winner there is a uh uh PDS and PRU that is the problem state uh the problem definition statement and uh requirement unit. Basically uh in a nutshell we go more into details of what exactly the n agency want and we and n agency share it with the startup more details more nitty-gritties and the startup MSME or the applicant they accept that we will be able to make it and then we declare the winner. So after declaration of winner we come to this stage uh correction we come to this stage the winner declaration. Once the winner is declared uh we uh begin with the process of documentation where we make the technical uh technical appraisal we make the project development budget and other documents. It takes approximately uh 20 days to a month and u uh uh lots of uh details go into we uh the project is divided into total five uh correction six milestones starting from zero. So the uh milestone zero is basically signing of contract from milestone 1 to milestone 3 is development of prototype. By milestone 3 we expect the prototype is totally developed. Milestone four is with respect to testing testing and certifications and milestone five is the user trial. So right now where I'm talking this is the milestone that I'm talking about milestone 5. After milestone 5 the when the uh uh SSD that is single stage composite test is over we progress towards the stage of A RFP and procurement. Uh so this is all the in a nutshell I have told about the cycle that we will be entering into. I will take you uh you you might have more questions. I'll take those those questions later. Basically this is the all the technologies key technology areas which we are working upon the other startups innovators that we are working with. These are the these are all the technologies that we are working into. uh the if I talk about the n agencies uh these are the n agencies that we are working with we have pan India we have total 28 partner incubators they are distributed all along the country and uh they their their their basic aim is to handhold the startup when uh we

believe that you are the expert in development you know how the technology will be developed however there are there are other aspects also where handholding is required. So that's uh where for Aditi we have different PIs. You can see those PIs are star marked. You can find these details in our website also. Uh the partner incubators for Aditi are uh uh they are not different but they are Aditi partner incubators are uh I I believe five or six partner incubators are for Aditi. For disk these are all 28 partner incubators are there. uh we have uh investor hub also which we call as II uh we have there are total 14 Indian investors that are there with the pool of total uh 100 uh 1100 cr and uh they are there to assist in gaining the matching contribution. Uh so this is the progress that we have uh we have had till now. We have total 28 uh partner incubators. We total number of challenges that the data is still Feb 2026. We have more numbers added up into this data. Uh in the month of March also we have had procurement. So data is still 31st. I stand corrected. The data is still 31st of January 2026. We have total 566 innovation challenges launched. Total applications that we have received so far is 13 uh 13 13,000. We have total 674 winners declared and we are working with total 25 agencies. Uh we have total 545 contracts signed and if I talk about uh the total a they are worth 3853 CR and total procurements that we have done is total 2229 CR. We have total 43 contracts procured and 57 a uh accorded. Uh this is the total and uh okay uh so this is this was a live last slide. Uh one more important aspect of the participation into IDEX challenge that I wanted to talk about is uh uh the grant is uh if you are participating in disk the grant is 1.5 maximum grant is 1.5 and uh if you are participating in Aditi the maximum grant is up to 25 CR uh we expect the startup or the participant to bring the matching contribution uh which which should be uh equal to or less than the correction which should be equal or more than the SPA grant. So for example just for example if your project budget is uh 2C cr for example your project budget is 2C cr so maximum grant that you can claim for is 1 CR and 1 CR matching contribution that you you will have to bring from your side in case of ADI for example your project budget comes out to be 40 CR. So 20 CR grant that you can claim for and uh 25 uh sorry 20 CR you have to bring the matching contribution. So this is how the uh partition is there and you can uh visit facts in our website for more details especially the financial facts before applying for the uh problem statements. So uh I have I'm I'm done with the uh presentation. You can ask your questions. Ma'am there is a question on uh yeah may I ask the question? >> Yes please. >> So one question is related to um actually while selecting any challenge it is very important to know the Q generally we are getting to know at the end of in in the duration of PDSPU. Is it possible we can get the Q first? you can write it to me. I'll uh uh I'll speak to the nodal officer whether at this stage it is required. You can uh write detailing that the with the reason why exactly you want moq at this stage. I'll uh speak to the n officer and reply back to you. >> Thank you. One more question. Um there is a process that only five challenges can be pursued by one company. >> Yes. So is there any change in this or uh >> no? >> Okay, thank you. >> Uh there is a chat. Uh can we apply for more than one challenge? You can apply for maximum five challenges. >> Ma'am, I had a question. I had previously applied for IDEX challenges and I did not receive any communication and uh one day like I used to check the portal too but one day I I logged into the portal I saw that I have been selected uh uh considered for HPSC uh and after that I saw not considered for further stage of evaluation. So but I did not receive any communication regarding that I have been considered for HPSC. So can >> no you when when you apply for uh when you participate in any challenge be it DIS OC or Aditi you give your email ID we communicate through your email id whenever you are shortlisted for the uh HPSC if in case we do not receive any acknowledgement if you have been shortlisted we do not

receive any acknowledgement we contact on the phone number as well. So I request all the innovators to be very careful with giving the email id as well as the contact numbers. We will be first we we send email and if we do not receive any acknowledgement we contact on phone also but it is very difficult to reach you out again once uh email ID and contact number is also not reachable. Uh are you talking about the one you applied for the open challenge? Uh no ma'am this is for uh dis uh okay as far as uh uh as far as army challenges are concerned there uh there is no disk challenge where innovator didn't turned up for the HBSC so maybe for some other knowled agency but army HPSC's all disk or Aditi participants have turned up for the HBSC so far okay ma'am hello ma'am Uh >> yes >> which has stated previous query applied in open challenge in army domain on November 2025. Here is my question is um does high high power selection committee is reviewed or not? My proposal. >> Yes. This is uh reviewed as in >> um I mean the proposal uh reviewing of uh HPS. No no no uh uh HPSC no uh review of proposal is a different thing. If you have applied for the open challenge the review of proposals happens at the PI stage. We uh we have to we from the 28 PIs we we choose three PIs and these partner incubators u review the OC I'm talking about OC only OC challenges but review of HPSC is not there >> all right uh in uh in my uh dashboard uh the status is reflecting in process does uh the rejection has happened or um can you throw some light on Yeah, basically what happens is uh if I if I talk about army challenges uh in OC we receive as much as 1,500 to,600 applications and out of them uh we receive uh nine or 10 applications for the HPSC. So uh it is difficult for for us to revert back to the startup with the the status that whether you you can wait and also OC uh the applications of OC happens in the batches because of the like like it it is very difficult for a nodal agency to uh go through 1600 applications their proposals and uh so what they do they divide them into batches. So for one OC we had total three batches of HPSC's. So uh then it becomes difficult for us to revert back to the startup whether to uh uh whether their application has been considered or not. What I can suggest you is you can uh um if you do not hear anything back after after 6 months for OC only I'm talking I'm not talking about this kind then you can reapply you can apply your proposals with more refined details maybe you would have missed out uh to some details you can also uh take advantage of all the defense events that happen around and meet with the uh uh officers of the nal agencies to understand more uh details that what is expected and what what solutions are they looking for? >> Ma'am, in terms of matching contribution, could you please explain once again uh is there any percentage that the startup should invest? Uh what's the ratio? >> Yeah. Yeah. Yes. So, uh the we the project development budget we call it PDB is divided into two parts. one is the grant the spark the the spark grant which is 1 m up up to 1.5 for disk and up to 25 cr for aditi and the rest uh is the matching contribution so for example your project development budget PDB is of 2C cr so one CR you can claim for uh from the uh spa the grant and one CR is the matching contribution the matching contribution has to be equal or more than a spark grant it can be just one rupee more or it can just be equal. >> Okay. So in terms Yeah. In terms of if the project valuation is like 41 cr then we claim 20 cr from ai and 21 from our end. Yes, you're right. >> Or is it? >> Yeah, basically matching contribution has to be slight the matching contribution has to be more than the uh grant amount. >> Some light on indicators uh which had uh which displayed in the dashboards regarding anything uh in whether it is an open challenge or disk or Aditi. Sorry I I I couldn't get your question. >> The status indicators uh I mean in process HPC uh the labels have been displayed right in the dashboard. >> You are telling me or asking? >> I'm asking ma'am. >> Uh okay. Uh yeah. So uh the IT team is working uh working upon that. uh we have progressed from

where there was no status update to now where it is it shows in progress. So the IT team is working upon that uh as soon as it is uh done then you will be able to uh see the status also >> another question. >> Yeah. Uh so sorry why just one uh so that matching contribution you mentioned if the project cost for disk itself is 1.5 cr then we can claim only like 75 lakhs right >> from the >> yes correct correct correct >> okay thank you >> and it can happen like for example your uh uh for example your for disk your project development budget is 4C cr so 1.5 you can claim from the grant and rest is the matching contribution. >> Okay. >> Ma'am, can the matching contribution be preunded? Can the company raise that um if the company would have raised that money already and uh can we still apply for the grant or during the grant will it be asked that again we have to raise that much amount of money? It's up to uh it's uh the ma the matching contribution is up to you. When we uh uh when we uh uh do the PDB uh when we get approval on PDB that time we ask you that from what resources you are getting your matching contribution. So that time you can say you can write that you are uh I mean it is uh from the innovator or it is from the investor you can uh get those details. >> Okay. Ma'am I have two questions. One is on the delivery timelines for these challenges. >> Okay. So ma uh so depending upon the uh challenge there are few challenges which are in which were there in Aditi 1 the timelines were just one year uh and there were few challenges which took approximately 24 months. Uh so it totally depends upon the problem statement. However, our uh normal, you know, the uh gestation period that normally we look for is for uh the most complex project also is 24 months. I have not seen a a timelines more than 24 months but again the project was very complex. So it's uh basically when you will be preparing the documents after this I'm talking after HPSC when you will be preparing the documents the nodal agency will be onboarded by then. So we go we go through it we scrutinize and then we understand that u uh what is the exact uh uh time period for the development of the prototype should be. >> Understood. And another question ma'am uh you also mentioned that uh company can bid for like know five challenges and uh so upper limit is five and company can get only one project or like you know if the uh technical proposal is good whether they will be allowed to another project or only one project is upper limit. If you do not have any idex challenge running as of now >> uh then you can go for total five projects in which only Aditi the limit for Aditi is one. You can have only one Aditi. You can apply but you can uh I mean you can run with only one Aditi project. You can apply in as many as possible. The limit is not for the application. The limit is for the uh uh running the project. It should not happen that you have two you can you cannot run two Aditi projects. >> Okay. >> So five limit is for the running projects. >> Okay. >> Got it. Thank you. Ma'am about disk. A disk has five. You you cannot have five uh more than five disc. Aditi has only one otherwise uh you can if if you already have projects running then then it will be it it should be a concern and also if you uh yeah also if you are uh uh somehow related to a company which already has IDEX projects that those projects also will be considered to yours I mean if you are uh u uh adviser with 51% of stakes over that company and they have running iodide project then also it is considered as your project only. I I hope I clarified this point. >> This is from Ninx. So already we are running two projects currently. >> They are disk OC or Adi >> um OC I think. Okay. So if you so if you already have two OC's you can again application is not a problem but once you are shortlisted you cannot have more than one Aditi and uh combination again you can have three projects and then again there is a combination that your projects that already are running here they should have if they are more than three three or more than three then they should have crossed milestone three. >> Okay ma'am. Yeah. Although at this stage I would

suggest you should apply as much in in as much challenges as possible and uh we can uh look at look at it later when we uh go for HBSC. >> Okay ma'am uh ma'am and then one more thing could you please tell me the following risk problem statement outreach datas? Uh yes you can ask. >> Uh PS1 risk PS1 uh uh right >> problem statement of development of AI module. >> Yes ma'am that you have mentioned earlier 17th April. >> 17th April hopefully >> evening hours. >> PS8 >> dis 8. >> Yes ma'am. Uh it will be hopefully on 27th of April. Uh uh for noon hours. >> Okay. Um disk 11 >> 10th of April uh evening hours, afternoon hours. >> Okay. Um >> ma'am what about 4 6 and 46 problem statements? >> Uh what I can uh do is um uh next time whenever there is uh an outreach uh uh just just give me a moment. So what I'll do is uh our next army's next outreach is on 27th. So on 27th I'll prepare a schedule of all the uh outreaches all the challenges and their outreaches and I'll uh flash it here so that uh uh you can prepare accordingly and also we do social media post also we you can keep following us and in social media post you will be uh you will come to know when is which challenges outreach is there >> on 27th what are the challenges you are covering ma'am? uh 27th we will be covering >> DG indigenous development of >> yeah we will be covering uh problem statement 8 of Aditi >> okay >> uh wait a moment we will be covering three disk challenges I I stand corrected uh we will be covering doing three disc challenges. They are disc uh they are from uh 11:13 and 14. >> 11:13. Okay, ma'am. Thank you. >> Yeah. Uh 27th uh our outreach will be from 2 p.m. to 5:00 p.m. >> Okay, ma'am. >> Thank you. >> Yeah, ma'am. I have a question regarding Sorry, regarding the matching contribution part. >> Yeah, tell me. >> So ma'am, you said that there is a program that helps startups to raise that funding. So is there a way that I could get in touch with them? >> Uh you can visit our website. There is uh in the website you will find there is a uh uh icon with the our partners. Yeah, our partners in that you will find IH. I is a small next I is capital H. The IDEX investors hub. you can find all the investors their details and you can get connected to them. However, uh once your uh challenge start uh once you are shortlisted as a winner then I I believe then the onboarding process starts. However, you can uh start getting in connection with them. Uh we have the details about them, their history and their contact details. You can go through it. Uh one more thing ma'am >> when we need to raise this funding round in which stage? Yeah. So uh uh right now we are at uh uh the outreach stage. Uh the application will close by 4th of May. Appro take approximately 1 and 1/2 month for the uh sorting of uh the uh winners and then the HPSC process will start. We have total 34 problem statements. So it will take time. As soon as you get the call for HBAC, you you do I mean you you you present your yourself in the HPSC and u you can take guess from that and after that you can start looking for the investors. The actual requirement for the investors uh or the investment part will come as soon as you sign the contract. As soon as you sign the contract the uh the milestone 0 closes in milestone 0 you have to give the Yeah. 10% of the total uh uh PDB and your part will be 5% of that 10. >> So ma'am we could raise it in stages right? >> Yes. Uh the uh matching contribution is to be uh given in stages and I stand corrected in milestone zero it is 10% contribution you have you will have to bring from your side 10% of the total PDB. >> Okay ma'am thank you. Ma'am all uh questions will be routed via gorio zoho mail.in or is it possible to get email id of the nodal officers? >> Uh uh getting email id of the nodal officer there is no problem I can share. However, the response uh will be delayed because for a nodal officer receiving an email which usually goes to spam also is uh they don't uh because lots of emails they do they they keep getting day in and out. So it will be better to forward it uh to the uh Gorov's ID and then we will revert uh we will forward it to the nodal officer because then it will be going from DIO. So I believe that's the better route. >> Thank

you. Yeah. Hello. Uh Sen here. >> Yes. >> Uh so I wanted to ask about uh the discussion for problem statement 7 uh under Aditi. >> So would that be happening today or is there a different uh session scheduled for it? >> We already had uh it is humanoid for combat and humanoid autonomous system. So we already had discussion with respect to this. Uh the recording will be uploaded uh on the YouTube channel. >> Okay. Okay. All right. Thank you very much. Sure. >> Uh ma'am quick question here. >> Yeah. for all the challenge that that we are talking about and the one which is your desk challenge 11 13 and 14 where do we get the details like what what exactly is the challenge or do we have to come into this uh 27th of March ch uh outreach program and then we will be able to know >> uh if you are talking about the problem statement the problem statements are there in the uh website so you can go to the website and there there will be a challenge uh uh logo you can click over that you will reach to the I mean if you want this 14 dis 14 then Aditi 4 you will see all the challenge problem statements all the node agencies problem statements will be there you can read the problem statement from there and uh once the recording is uploaded uh it will be timestamped so through with the help of time stamp you can directly uh in the YouTube recording you can go to the specific state Are there any uh more questions? We need a disc outreach date for 13 25 and 26. for uh 25 and 26 will be 13th of April. >> 13th of Okay. Uh and then 13 ma'am please >> 13 will be on 27th of April. I suggest 27th of April please uh for a while or you can look at the recording also. We'll flash the uh schedule for the outreaches. However, please be informed that these uh schedules we try as much as possible to stick to the to the schedules but uh nonavailability of nal officer for any reason may cause the uh reschedule of the specific uh problem statement and outreach. >> Okay, thank you. >> We will be having general outreach uh almost all the days. You can keep following us on the social media. you will come to know uh all the general outreaches are more or less same only they will be recorded and they will be uploaded also uh for the specific following statements from the other knowledge agencies also you can follow the social media uh it will be uh uh uh there in the social media >> okay ma'am >> yeah so if there are no more questions u uh we will meet again on 27th of March for the outreach of uh next set of challenges I'll again uh reiterate the challenges that we will be conducting on 27th. On 27th uh from 2 p.m. to 5:00 p.m. we will be conducting the technical outreach of disk problem statement 11 that is integrated indigenous development of germanium lenses for thermal imagers. uh disk problem statement number 13 that is indigenous development of MFR of DEA detector cooler assembly used in HHTI with LRF and HHTI bicolors and also we will be conducting outreach for problem statement number 14 that is indigenous development of MFR for bio volometer based uncool infrared focal plane array FPS these three problem statements will be discussed from 2 p.m. to 5:00 p.m. on 27th. Okay. Thank you ladies and gentlemen for joining the uh outreach. Uh we will meet on 27th. Thank you so much. >> Thank you.